Disease Occurrence Data

APHS 20203 • Epidemiology & Biostatistics

Learning Objectives

By the end of this week, you should be able to:

1. Identify key factors that influence **epidemiologic data quality**.
2. Distinguish between **vital statistics** and **reportable/notifiable disease data**.
3. Identify major **surveillance programs in the United States**.
4. Describe major **sources of epidemiologic data** used in public health.
5. Explain the role of **international organizations** in collecting and disseminating epidemiologic data.

Review of Mortality: Real Data (CDC NVSS, 2023)

- Source: CDC NCHS Data Brief 521
- Selected findings:
  - Life expectancy: **78.4 years** (↑ 0.9 from 2022)
  - Age-adjusted death rate: ↓ **6.0%** (798.8 → 750.5 per 100,000)
  - Age-specific death rates decreased for all age groups ≥5 years.
  - Leading causes unchanged: heart disease, cancer, injuries.
  - Infant mortality: **560.2 per 100,000 live births** (no change).

Leading Cause ≠ Highest Risk

- **Leading causes of death** are ranked by **proportion of total deaths (PMR)**.
- Ranking answers “**how many deaths?**” → **Burden**.
- **Risk** answers “**how likely is death?**” → measured by rates (e.g., **CSMR**).
- A cause with a lower rank can still pose **high risk** to specific populations.
- Public health concern depends on **context**, not rank alone.

> **Burden ≠ Risk**

Example: Suicide

- Suicide is **not a top-ranked cause of death overall**.
- But it is a **leading cause of death among young adults**.
- This reflects **high cause-specific mortality rates** in certain groups.
- Key contrast:
  - PMR (overall): low
  - CSMR (subgroups): high

Conceptual Transition

- You just interpreted multiple mortality measures using one short summary.
- **Question:** What assumptions are we making about the data behind these numbers?
- *(Write down your responses; we will revisit soon.)*

Epidemiologic Data

- Epidemiologic data are **population-based data** used to:
  - Describe disease occurrence
  - Monitor health trends
  - Identify disparities
- Epidemiologic data capture:
  - **Morbidity** (disease, illness)
  - **Mortality** (death)
- Major sources of epidemiologic data:
  - Vital statistics
  - Reportable/notifiable disease data
  - Survey and surveillance data

Factors Affecting Epidemiologic Data Quality

- Data quality reflects whether the assumptions behind our measures hold.
- Key dimensions:
  - **Coverage:** Who is included?
  - **Accuracy:** Are events classified correctly?
  - **Consistency:** Are the same definitions used over time and place?
  - **Timeliness:** How current are the data?
  - **Processing and quality control:** Errors and checks

Two Types of Epidemiologic Data Systems

Epidemiologic data systems serve **different public health purposes**.

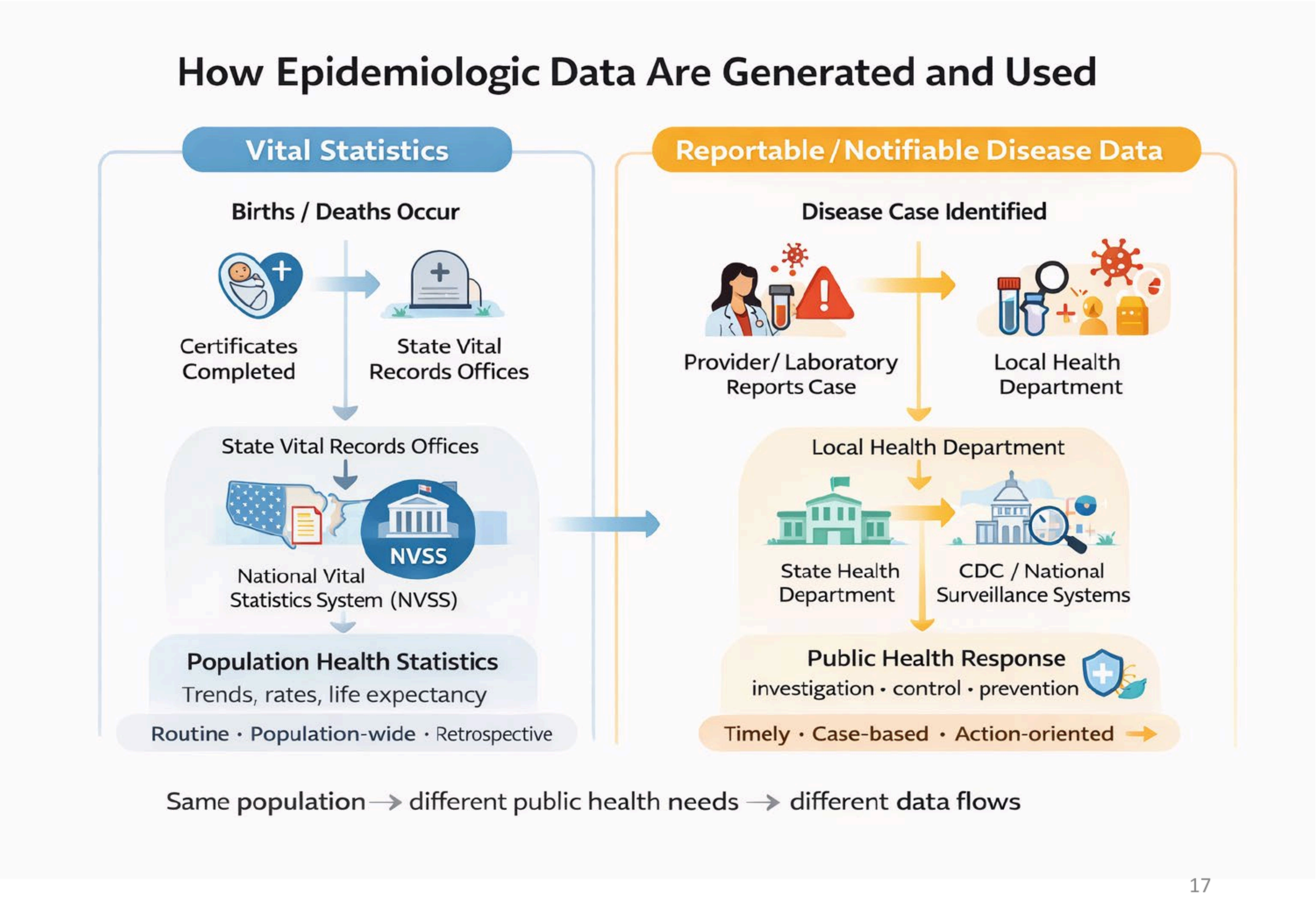
VITAL STATISTICS

- Track *life events* (births, deaths)
- Population-wide and complete
- Designed for **long-term patterns**

REPORTABLE/NOTIFIABLE DISEASE DATA

- Track *specific diseases*
- Focus on timeliness and rapid detection
- Designed for **public health action**

> Same population → different questions → different systems


How Epidemiologic Data Are Generated and Used
Vital Statistics
Births / Deaths Occur
Certificates Completed
State Vital Records Offices
State Vital Records Offices
NVSS
National Vital Statistics System (NVSS)
Population Health Statistics
Trends, rates, life expectancy
Routine · Population-wide · Retrospective
Reportable / Notifiable Disease Data
Disease Case Identified
Provider/ Laboratory Reports Case
Local Health Department
Local Health Department
State Health Department
CDC / National Surveillance Systems
Public Health Response
investigation · control · prevention
Timely · Case-based · Action-oriented
Same population → different public health needs → different data flows
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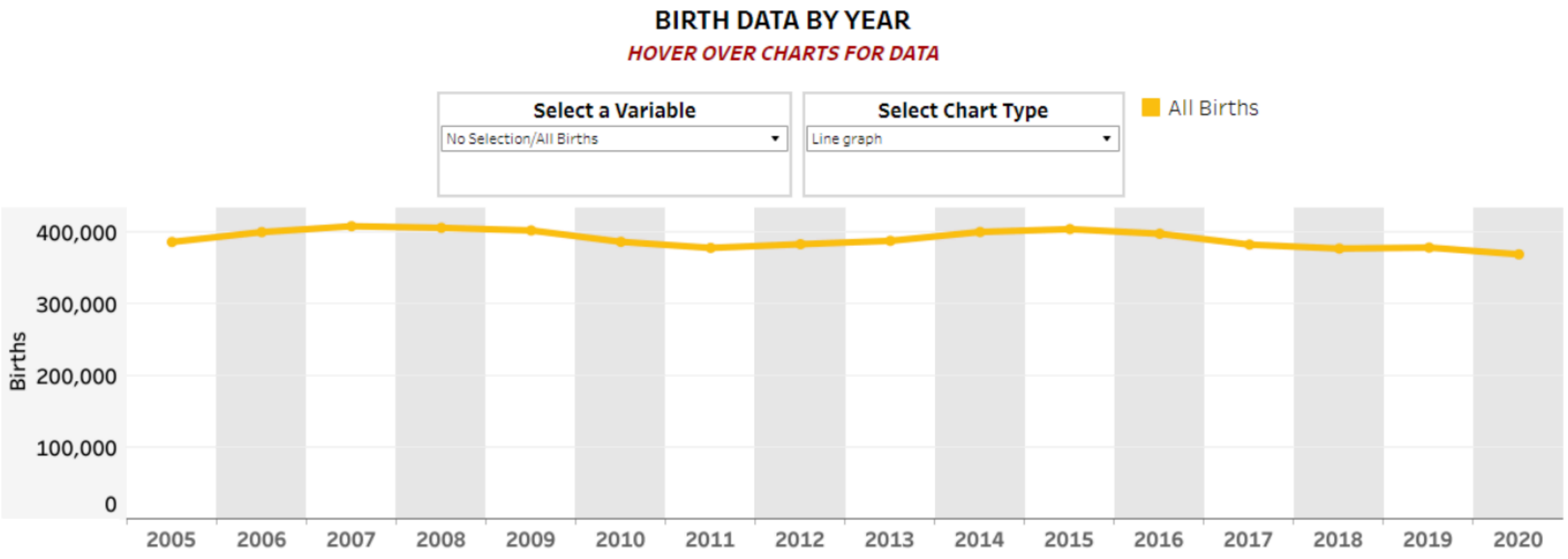
Vital Registration System

- **Vital events:** births, deaths, fetal deaths, marriages, divorces
- In the United States, these events are compiled through the **National Vital Statistics System (NVSS)**.
- https://www.cdc.gov/nchs/nvss/index.htm

Information Collected by Birth and Fetal Death Certificates

Variable	U.S. Certificate of Live Birth	U.S. Report of Fetal Death
Name	Child's name	Name of fetus (optional)
Disposition (e.g., burial)	Not applicable	Disposition of fetus
Location	Facility name	Where delivered
Mother identifying information	Name, age, and place of residence	Name, age, and place of residence
Mother	Height, weight, number of previous live births	Height, weight, number of previous live births
Conditions contributing to fetal death	Not applicable	Initiating cause and other causes (e.g., pregnancy complications, fetal anomalies)
Risk factors in pregnancy	Diabetes, hypertension, infections	Diabetes, hypertension, infections
Congenital anomalies	Anencephaly, cleft palate, Down syndrome	Anencephaly, cleft palate, Down syndrome
Father	Name and age	Name and age

Vital Statistics Example: Texas Birth Data

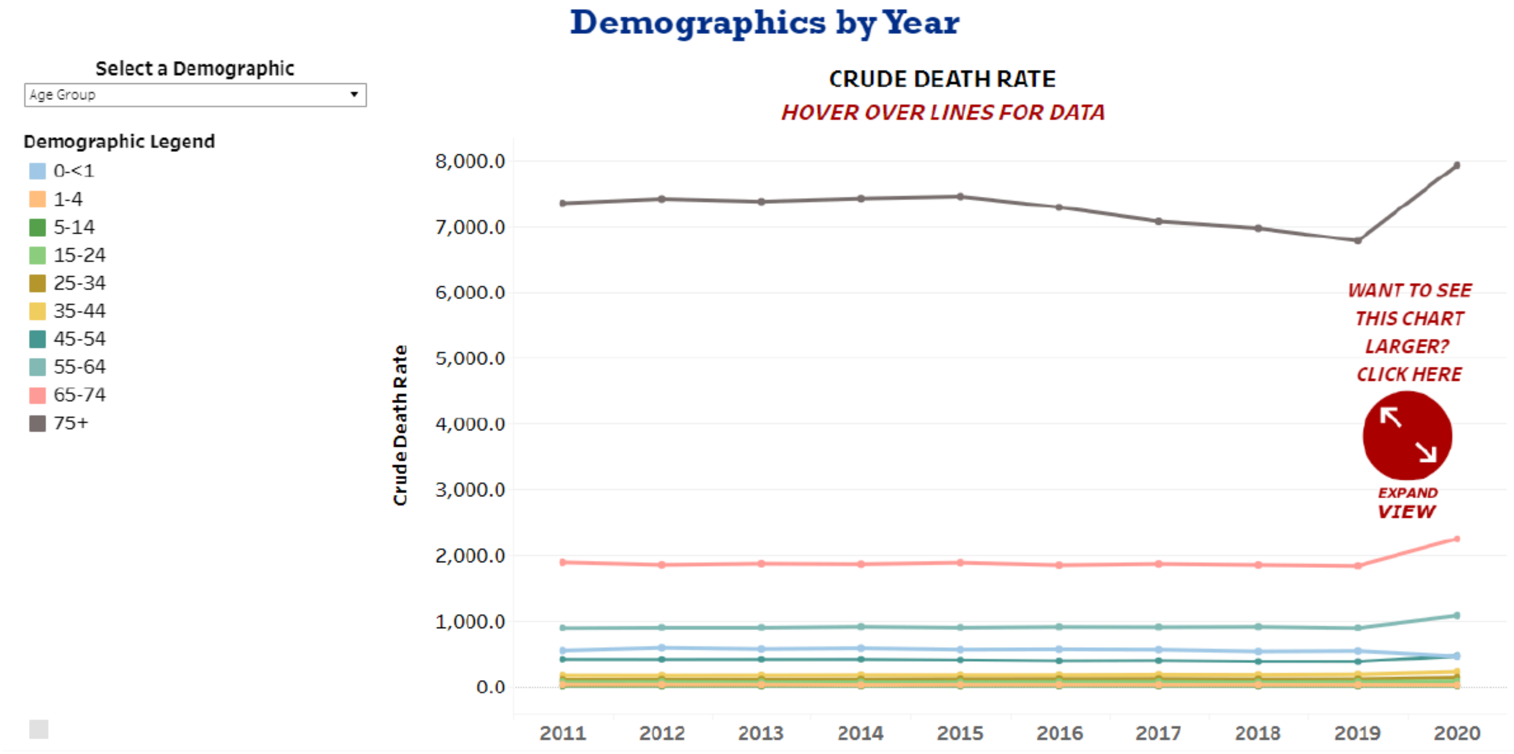


[Source: https://healthdata.dshs.texas.gov/dashboard/births-and-deaths/live-births](https://healthdata.dshs.texas.gov/dashboard/births-and-deaths/live-births)

Discussion

- What do you notice about this trend?
- Which features of the vital registration system make this kind of trend analysis possible?

Vital Statistics Example: Texas Death Data



https://healthdata.dshs.texas.gov/dashboard/births-and-deaths/deaths

Discussion

- What do you notice when deaths are shown by age group?
- Why is this *crude death rate* shown separately by age group?
- Which mortality measures from earlier in class rely on this same death data? 21

Why Do Some Diseases Have to Be Reported?

- Some diseases require **immediate public health action**.
- Waiting for routine statistics is too slow.
- These data support **detection, response, and control**.

> Different purpose → different data system

Reportable/Notifiable Disease Data

- **Case-based surveillance**
- Focus on **specific diseases** of public health importance
- Information typically includes:
  - Case counts (incidence)
  - Basic person, place, and time
  - Key clinical outcomes
- Key tradeoff:
  - Strength: **Timeliness**
  - Limitation: **Incomplete population coverage**
- Designed for action, not completeness.

Reportable vs. Notifiable — What's the Difference?

REPORTABLE DISEASES

- Reported to **local or state** health departments
- Used for **local public health action**
- Lists vary by jurisdiction

NOTIFIABLE DISEASES

- Reported to **national authorities** (e.g., CDC)
- Aggregated for **national monitoring**
- De-identified (age, sex, location)

- Every notifiable disease is first reported locally.

> Same case → different reporting level → different purpose

Examples of Nationally Notifiable Diseases

National Notifiable Diseases Surveillance System (NNDSS): the endpoint of the data-report flow

- Anthrax
- COVID-19
- Botulism
- Gonorrhea
- Hepatitis A, hepatitis B, hepatitis C
- Human immunodeficiency virus (HIV) infection
- Meningococcal disease
- Mumps
- Syphilis
- Tuberculosis
- Viral hemorrhagic fever (includes Ebola virus)
- Zika virus disease

Source: Centers for Disease Control and Prevention. National Notifiable Diseases Surveillance System (NNDSS). 2022 Nationally Notifiable Conditions. Available at: https://ndc.services.cdc.gov/event-code-list-other-surveillance-resources/. Accessed February 20, 2025.

Discussion: Nationally Notifiable Diseases

1. Pick one disease on the list. At what level would it first be reported?
2. Why would CDC want to track this nationally?
3. Would birth or death data ever appear on this list? Why not?

Surveillance: From Reporting to Monitoring

- Reporting captures **individual cases**.
- Surveillance looks for **patterns over time**.
- Goal: inform **ongoing public health action**.

> Reporting → Surveillance → Action

Two Approaches to Public Health Surveillance

Public Health Surveillance	Syndromic Surveillance
Routine, ongoing monitoring	Real-time, early warning
Uses confirmed diagnoses	Uses symptoms and signals
Tracks incidence and trends	Detects unusual patterns
Usually takes place in: Public health departments, registries, national systems	**Usually takes place in:** Emergency departments, urgent care, pharmacies
Example: flu incidence over years	Example: spikes in fever-related ER visits

Exercise: Types of Surveillance

Match each scenario with the correct type:

PUBLIC HEALTH SURVEILLANCE

Long-term, systematic, incidence and policy focus

SYNDROMIC SURVEILLANCE

Real-time, early warning, symptom and pattern focus

Exercise: Types of Surveillance — Scenarios

1. Health officials analyze flu vaccination rates every year to guide next season’s immunization campaign.
2. A sudden rise in ER visits for fever and cough signals a possible flu outbreak.
3. Monitoring long-term cancer incidence to plan screening programs.
4. Tracking over-the-counter cold medicine sales during winter to catch early spikes.

Focus on *why* one approach fits better than the other.

Public Health Surveillance Programs

- **Communicable and infectious diseases**
  - Track cases, hospitalizations, and spread
  - *Example:* COVID-19 surveillance (cases, hospitalizations)
  - *Example:* National Wastewater Surveillance System (NWSS)
- **Chronic diseases**
  - Monitor long-term incidence and outcomes
  - *Example:* Cancer registries (incidence, types, outcomes)
  - *Example:* CDC Chronic Disease Indicators (CDI; state-level indicators for chronic disease patterns)
- **Risk factors for chronic disease**
  - Track behaviors and exposures
  - *Example:* BRFSS (obesity, smoking, physical activity)

National Center for Health Statistics (NCHS)

NCHS is a major source of U.S. population-based health data.

- NCHS data complement surveillance by describing population health beyond reported cases.
- **Survey-based systems**
  - **NHIS:** health status, access to care, prevention
  - **NHANES:** exams and interviews (diet, activity, disease prevalence)
- **Vital statistics**
  - **NVSS:** births, deaths, life expectancy, leading causes of death

Why International Organizations Matter

- Diseases cross national borders.
- Data must be **comparable across countries**.
- Surveillance and response require **coordination**.

What International Organizations Do

- **Surveillance and reporting** — global monitoring
  - *Example:* WHO Global Influenza Surveillance and Response System (GISRS)
- **Policy guidance** — shared recommendations
  - *Example:* UNAIDS guidance for HIV/AIDS programs
- **Resource support** — capacity and access
  - *Example:* Gavi, the Vaccine Alliance

Key Takeaways: Epidemiologic Data

- Epidemiology depends on **population-based data** to describe health patterns.
- **Different public health questions require different data systems.**
- **Vital statistics** describe long-term population patterns (births, deaths).
- **Reportable disease data** support rapid detection and public health response.
- **Surveillance** turns reported data into ongoing monitoring and action.
- **Data quality** determines what conclusions can (and cannot) be drawn.
- Selecting an appropriate data source is a **critical epidemiologic decision**.